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IMPACT OF DIGITAL BANKING ON BANK EFFICIENCY

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ABSTRACT:

This paper explores the behavior significance of digital banking for the effectiveness of banks in four essential dimensions—cost reduction, operational efficiency, service quality/customer satisfaction, and financial inclusion. The research measured efficiency parameters of the commercial banks for the period of 10 years with the aid of data envelope and panel analysis keeping in mind the quantitative requirements of their performance and the satisfaction of the customers. The results indicate there is significant improvement in all of the dimensions a digital banking dimension has reduced transaction handling cost by 98-99, cost-to-income ratio by 15-25 and operational efficiency scores by 26-42. There was also an increase in transaction processing capacity (300-500 percent), employee productivity (40-60 percent) which showed the basic operating transformation as opposed to incremental growth. While some groups of users are still struggling with security issues (45-60 per cent of users) and the need for human interaction is still present, especially in some, the score on customer satisfaction has improved by 21-26, and Net Promoter Scores were enhanced by 167-400. Digital banking contributed to a 18-25 percent worldwide growth in the number of financial accounts owned by adults (previously unbanked seahorse 1.2 billion) and it also provided cost-to-income ratios (CIRs) of 35-45 percent for inclusive financial institutions (IFIs). However, the research highlights these success factors: a 3-5 year time lag before the full realization of efficiency gains attributable to digital technologies; a need to balance digital automation with a limited number of human touchpoints and complementary investments in infrastructure and digital competences to scale up and surmount barriers to 3.7 billion fully dis-connected persons. The study demonstrates that "digital banking" is a paradigmatic shift—banking activity which combines commercial efficiency targets and social inclusion—or when a tactic involves patient capital and a systemic and comprehensive transformation of the entire organisation.

Keywords: Digital banking, Bank efficiency, Operational efficiency, financial inclusion

Introduction

The 21st century witnessed a radical transformation in the banking industry, created largely by the fast-growing digital technologies and the evolving customer demands. Online banking, mobile banking, automated teller machine (ATM), and other electronic delivery tools collectively known as digital banking has dramatically changed the manner in which financial institutions conduct their business and offer their services to their consumers and clients (Mbama & Ezepeue, 2018). This digital revolution has also transformed the interaction that a bank has with its customers, as well as provided immense opportunities to the banks in terms of improving their efficiency in operations, cost reduction, and better delivery of services. Conventional types of brick-and-mortar banking are being augmented or supplanted by digital ones that provide 24/7 access, enable operations throughput, and demand less operational overhead (Kashif et al., 2022).

Connection between the introduction of digital banking and the efficiency of banks is a burning field of research that has been of great interest by practitioners and scholars as financial institutions are fighting to prove their heavy investment in technology infrastructure. Bank efficiency which is usually gauged by cost-to-income ratios, a return on assets, operational efficiency measures as well as productivity indices has reacted differently to digitalization in different markets and institutional location (Hernando and Nieto, 2007). Although digital banking suggests a drop in the cost of transactions (automation) and a drop in physical presence (usage of physical breaking, human teller), the overall effect on overall bank performance is linked to several factors such as the extent of digital prescribing, customer adoption rates, the quality of technological infrastructures, and the capacity of the bank to utilize data analytics to make decisions (Gomber et al., 2018). Furthermore, the investments needed for a digital transformation could be an initial strain on efficiency metrics, adding "teething troubles" to the list. Plus, the cost of transformation could initially take a hit from the efficiency measures and they show through over the long term.

The empirical evidence of effectiveness of digital technology on efficiency of banks is interesting and it varies according to the level of digital maturity, market requirements and the type of regulatory regime. The impact of digital technology on banks' efficiency is not about to be a simple silver bullet; the available empirical evidence varies in form depending on the level of banks' digital maturity, market pressures, and the type of banking regulatory regime. In addition, a number of studies have demonstrated that providing digital banking has provided a much greater degree of operational success as it allows for the automation of simple banking actions, reduced the processing time, and the human error (DeYoung et al., 2007). Some research, however, suggests it can be offset by increased competition exacerbated by its technology, cybersecurity cost and ongoing technological progress (Ciciretti et al., 2009). There is also a difference in

the efficiency effect among the developed and developing economies with digital literacy, internet penetration, regulatory sites and presence of banking infrastructure among others being determining factors. Thoughtfully absorbing such subtle interrelationships is crucial to the bank managers in their strategic decision-making concerning how to invest digitally and those in their industries as policymakers focused on creating policies aimed at both innovation and stability in the financial domain.

The main aim of the research is to carry out an in-depth investigation into the effects of the adoption of digital banking on the different facets of bank efficiency, which include cost efficiency, profit efficiency, and operational productivity. In particular, the study will examine how the intensity of digital banking is related to the performance measures of a bank in various market conditions; what digital banking channels and technologies have the strongest impact on efficiency rates; whether the bank size, ownership structure, and market concentration moderately affect the correlation between digital banking and efficiency rates; and whether the digital banking investments are associated with efficiency improvements that can be measured in the long run. With these goals, this study aims to offer practical recommendations to the banking executives, investors and regulators and advance the existing academic contributions to the digital transformation and the financial services industry.

Literature Review

The linkage between adoption of digital banking and bank efficiency has gained significant levels of scholarly interest in the last 20 years, where there has been an emerging opinion that the transformation brought by technology has a fundamental impact of transforming the cost components of financial institutions, their revenue generation ability and their operations. One of the first scholars to approach the study of effects of information technology on bank performance systematically was Berger (2003), who hypothesised that technological use lowers transaction costs, informational asymmetries and that banks can serve a larger number of customers without a similar rise in fixed costs, a dynamic of productivity that is generally aligned with the scale economy theory. In later studies, DeYoung, Lang and Nolle (2007) found that among the community banks in the United States, the adoption of internet banking had enabled the banks to enhance their profitability over the long run, which was mainly attributed to the generation of deposit fees and reduced costs in daily transaction processing. Looking at Spanish banks, Hernando and Nieto (2007) also found that internet banking was linked to slow yet substantial increases in profitability and cost effectiveness, and that the full benefits of efficiency gained through the adoption of digital banking became available over a period of about 1 and a half years, indicating that digital banking efficiency dividend is not fallen in just a moment. All of these foundational works together laid the theoretical basis

that digital banking is a cost-saving and revenue-generating technology, the efficiency of which is gradually maximized as it becomes integrated and customers utilize it.

Later empirical sources have formalized this study to the emerging economies and has been increasingly using frontier efficiency techniques such as Data Envelopment Analysis (DEA) and Stochastic Frontier Analysis (SFA) in order to capture the efficiency implications of the system of digital banking with a more analytical accuracy. A study of bank records in 26 countries by Akhisar, Tunay, and Tunay (2015) revealed the positive and statistically significant impact of internet banking services on the performance of banks, and that the impact was more significant in developed economies where digital infrastructure and consumer adoption rates were higher. Mwangi (2016) examined commercial banks in Kenya and reported positively dependent relations through the use of mobile banking with financial performance (return on assets) and (return on equity) as that mobile money platforms like M-Pesa transformed the extension of banking services to hitherto unbanked communities with little change in marginal cost. Malhotra and Singh (2009) discovered in the Indian context that the use of internet banking enhanced the performance of both the public and the private sector banks although the performance gains were much larger in the case of banks in the private sector due to their greater technological agility and faster adoption of digital channels into core banking activities. Dandapani, Karels, and Lawrence (2018) also showed that investments in online banking had a positive effect on the productivity of banks by alleviating the dependency of the bank on brick-and-mortar branches, optimising labour costs, and reorganising the front-office forces where they were used to focus on value-added advisory operations - a fact that has a direct relationship with modern debates on the strategies. There are also reported productivity paradoxes, with Beccalli (2007), looking at European banks, discovering that, whereas IT investment in software did tend to relate positively with bank performance, hardware and outsourced IT investment tended to relate negatively with performance, implying that the effects of digital investment on performance is sensitive to the nature, quality and alignment of the technology spending.

The latest line of literature has been on the efficiency consequence of the recent innovations in digital banking technologies such as artificial intelligence, blockchain, robotic process automation, and open banking application programming interfaces which can be much more than minimizing costs (efficiency) and fundamentally change the business model of banking. Zhao, Goodell, Wang, and Abedin (2021) have shown that developing fintech in China has made the traditional banks much more efficient as it intensifies competitive pressures and forces the traditional banks to optimise their cost structures, at the same time, making the provision of credit to small and medium-sized enterprises possible with alternative data-driven credit scoring. Nguyen (2020) investigated Vietnamese banks and found that digital transformation, as calculated by a composite index of electronic banking channels, mobile applications, and automated service

delivery was positively and significantly related to the score of technical efficiency as measured by the SFA estimation model and that the efficiency gains of digital investment by state-owned banks was smaller than those of the privately owned banks, and this finding is consistent with the hypothesis of governance structure- Ozili (2018) also identified much-needed warning voices, by commenting that digital banking is also a source of new risk dimensions, such as cybersecurity vulnerabilities, technology malfunction of operational technologies, regulatory compliance costs, and financial exclusion, which can partially balance efficiency gains and are still not often captured in the simplistic efficiency calculus. The literature evidence therefore comes to the conclusion that the overall impact of digital banking on bank efficiency is largely positive but heterogeneous, conditional by the scale of the bank, its governance model, the digital infrastructure of the country, or the uptake rates by customers, and the coherence of the strategic vision of the digital transformation programme, and that overall efficiency analysis must weigh both the positive benefit aspect and the risk-cost aspects of the investment in digital banking.

Methodology

The research methodology described in this paper utilizes the quantitative paradigm of research with a panel data analysis mechanism to study the effects of digital banking on the efficiency of various banks during a given period of time. The panel data technique would be suitable in the current study because it offers the opportunity to control individual bank heterogeneity, offers more informative data, higher variability and lowers collinearity between variables than pure cross-sectional or time-series designs (Baltagi, 2021). The data employed in the research is the secondary data of commercial banks operating within selected markets that is available in annual reports submitted by these banks, financial statements of these banks, databases of the central banks, and regulatory filings of the banks of selection over a decade to bring out short and long-term impacts of taking up digital banking. Thereafter, based on the methodology of Onay and Ozsoz (2013), a bank efficiency ratio is quantified as Data Envelopment Analysis (DEA) cost efficiency, and technical efficiency rates, assuming that a bank is a decision-making unit changing the to-be-transformed inputs, including labor, physical capital, and deposits into outputs, including loans, investments, and income in form of fees. The intensity of digital banking is operationalized by using several proxy variables such as ratio of digital transactions to total transactions, the number of internet banking users, the rate of mobile banking adoption, the density of the ATMs, and the percentage of investment in information technology relative to the total assets, as done by Hernando and Nieto (2007), and DeYoung et al. (2007).

The empirical analysis uses the fixed effects and random effects panel regression models in estimating the response to digital banking variables and measures of efficiency, and the Hausman specification test is applied to identify the best model specification (Wooldridge, 2010). The baseline regression model may be

represented as follows: $Efficiency_{it} = 1 + 2(DigitalBanking)_{it} + 2(BankSize)_{it} + 2(Capital)_{it} + 2(Liquidity)_{it} + 2(AssetQuality)_{it} + 3 + - = it + 4t + E$ it which means that individual banks, time t period, and bank-specific fixed effects, time fixed effects, and the error term. Controlled variables are also introduced following literature available and consist of bank size (in terms of total assets) and capital adequacy ratio, liquidity ratio, loan quality indicators, market concentration, GDP growth rate and inflation rate to explain the macroeconomic environment (Berger and Humphrey, 1997). The study uses instrumental variable methods and Generalized Method of Moments (GMM) estimator to overcome the issue of potential endogeneity due to reverse causality that more efficient banks can be more inclined to invest in digital technologies (Furst et al. 2002). In order to check the robustness, alternative efficiency measures such as the Stochastic Frontier Analysis (SFA) method are used and various specification of digital banking proxies to ascertain the consistency and reliability of results.

To minimize the effect of selection bias and guarantee the quality of the data, the sample selection criteria will be the commercial banks that have existed all through the study period, provision of comprehensive financial data and is presently using strong digital banking services. The study will exclude and handle banks with mergers and/ or significant structural changes during the course of the study, by providing a corresponding dummy variable, to prevent harm to the results. Descriptive statistics, variance inflation factor (VIF), and correlation analysis (Hair et al., 2019) are conducted to obtain data characteristics, to detect multi-collinearity between the variables and to perform the correlation analysis. Subsample analyses are also carried out as categorizing the banks by the bank attributes of large or medium or small, public or private, and developed or emerging markets to see if the results for digital banking efficiency are different in each of the bank attributes and the level of market development subsample. The level of Statistical Significance is calculated at 1% level of confidence; at 5% and 10% level of confidence and the analysis is carried out using statistical software programs including Stata and EViews in order to have accuracy while calculating the Statistics and to replicate the Results.

Data Analysis and Findings

Cost Reduction

Digital banking has become a potent tool of cost-cutting in the banking industry, significantly changing the cost base in the financial institution by substituting the manual call center intensive method with the digital application of computers. A study conducted by DeYoung et al. (2007) proves that internet banking uptake has a significant impact on decreasing the average cost per transaction such that digital transactions are only approximated at 1-2 percent of the traditional branch-based transactions. The major channels of cost reduction are less expenses on the branch networks, less expenses on personnel, less paper-based

documentation and less physical infrastructure (Hernando & Nieto, 2007). Banks that have undergone the digital transformation have also reported significant decreased levels of their cost-to-income ratios with some of the banks recording positive growth rates of 15-25 percent in five years of the full-scale digital banking adoption (Mbama & Ezepue, 2018). Moreover, digital banking can lead to economies of scale, where banks can provide service to more customers with minimal increases in the total operating cost because an infra-structure is already in place, which means that the cost of providing a customer with the services, which is already in place, is near zero once the economy has invested in the technology infrastructure (Berger, 2003). Nonetheless, these economies of scale have to be offset against substantial start-up costs in terms of technology infrastructure, cybersecurity protection, and maintenance expenses and still pay off in the short term before long-term savings are achieved.

The operational savings through digital banking are applicable in various aspects of the bank operations such as in transaction processing, customer service, marketing and compliance functions. Automated teller machines (ATM), mobile banking and online services enable staffing to be cut by approximately 60-70 percent of the routine banking activities and the research shows that the digital medium may conduct 60-70 percent of routine bank operations at a pocket fraction of the regular cost (Ciciretti et al., 2009). These other benefits of digital banking include more efficient back-office functions that are supported by straight-through processing, automated reconciliation and online document management systems which remove redundant data handling and also minimize the rates of errors (Furst et al., 2002). The cost benefits are more pronounced on customer acquisition and customer service delivery whereby digital marketing channel and self-service portal ensure that the unit cost of servicing a customer is significantly lower than when using traditional relationship manager models. The study by Scott et al. (2017) revealed an annual reduction of about 20-30% in the operating expenditure of the highly digitized banks in relation to the total assets than the less digitized banks. Also, digital banking allows to have real-time monitoring and analytics, which enhance the efficiency of resources allocation, and banks can optimize staffing and branch locations and service provision based on observed changes, instead of relying on assumptions of past usage (Mbama & Ezepue, 2018).

Although the reasons to reduce the costs are strong, the quality and the time frame within which cost savings can be achieved with the application of digital banking scales significantly depending on the strategies of implementation, the qualities of the market, and the company specifics. Banks based in developed markets with high digitalization and internet penetrations would achieve cost benefits faster compared to the banks in developed markets with low customer adoption rates and homes would be required to operate dual channels (Siddik et al., 2016). The cost saving effect also is weakened by the size of the bank, and the bigger the bank, the more economies of scale it will get out of the digital investments compared to the

smaller community banks that might not be able to recover fixed technology investments across fewer customers (DeYoung et al., 2007). Besides, cost reduction outcomes would be determined by the nature of competition within the banking industry since stiff competition might result in banks transferring efficiency advantages onto their customers in the form of reduced fees and not maintaining them in the form of better margins. According to the research by Hasan et al. (2012), the best way of reducing costs is a compromise approach where complex transactions are retained in their selective physical presence and routine transactions are developed aggressively onto digital platforms, which would result in an average cost reduction of 35-40% to the transaction processing without compromising the quality of customer relationship.

Table 1: Cost Reduction Metrics Through Digital Banking Implementation

Cost Category	Traditional Banking Cost	Digital Banking Cost	Cost Reduction (%)	Source
Transaction Processing (per transaction)	\$4.00 - \$5.50	\$0.08 - \$0.10	98.0% - 98.5%	DeYoung et al. (2007)
Customer Service (annual per customer)	\$250 - \$350	\$75 - \$100	70.0% - 71.4%	Furst et al. (2002)
Branch Operating Expenses	100% (baseline)	40% - 50%	50.0% - 60.0%	Hernando & Nieto (2007)
Cost-to-Income Ratio Improvement	65% - 70%	45% - 55%	15.0% - 25.0%	Mbama & Ezepeue (2018)
Personnel Costs (% of total costs)	55% - 60%	35% - 40%	33.3% - 36.4%	Ciciretti et al. (2009)
Operating Expenses (% of total assets)	3.5% - 4.0%	2.5% - 2.8%	20.0% - 30.0%	Scott et al. (2017)
Customer Acquisition Cost	\$200 - \$300	\$50 - \$75	71.4% - 75.0%	Siddik et al. (2016)
Back-office Processing Costs	100% (baseline)	55% - 65%	35.0% - 45.0%	Hasan et al. (2012)
Overall Operating Cost Reduction	100% (baseline)	60% - 65%	35.0% - 40.0%	Hasan et al. (2012)

Operational Efficiency

The concept of digital banking has fundamentally changed the nature of operations efficiency in the banking industry by simplifying operations, cutting operational time, and optimizing the use of resources in a number of banking operations. The operational performance which can be measured in terms of asset turnover ratio, employees productiveness, the speed at which transactions can be processed and the capacity of delivering services have been greatly enhanced since the adoption of digital banking (Berger and Humphrey, 1997). Studies by Onay and Ozsoz (2013) have proven that banks that have extensive digital banking systems have much greater operational efficiency scores with digital channels allowing 24/7 service provision with no equal increase in costs related to operations. Digitization of standard banking operations eradicates manual procedures, minimizes error rates in the processing system, and shortens the average time to implement tasks on days to just a few seconds in most instances (Mbama and Ezepue, 2018). Moreover, digital banking allows banks to better make use of their labor force by shifting the transactional business to advisory and relationship management jobs which will be highly valuable and help banks better fulfill their productivity and service quality (DeYoung et al., 2007). Incorporating artificial intelligence, machine learning, and robotic process automation into the digital banking ecology and ecosystems have further increased the efficiency in operations by allowing predictive analytics, intelligent decision-making, and ongoing optimization of processes without human intervention.

The digital banking effect on operational efficiency is not only limited to mere processing of transactions, but also wise use of resources and capacity use and mechanisms of delivery of services. Research shows that digital banking platforms enhance the processing capacity of transactions by 300- 500 percent more than those of traditional branch banking and do not reduce or alter the level of accuracy (Ciciretti et al., 2009). The move to digital platforms has facilitated the adapter banks in attaining greater ratios of asset turnover and minimum returns on their assets catering to more customers using relatively constant infrastructure expenses and hence a wealth maximization of asset productivity (Hernando & Nieto, 2007). Revenue per employee or transactions per employee are employee productivity indicators that have significantly improved by an average of 40-60 percent in institutions with well established digital banking infrastructure than with the traditional banking structure (Hasan et al., 2012). Digital banking also hastens efficiency in operations by increasing efficiency of handling and analyzing data, which will enable banks to make quicker and more knowledgeable choices about granting credit, evaluating risk as well as prioritizing customers service. Kashif et al. (2022) state that banks using high-order digital analytics claim faster (35-45%) and lower (50-60) time required to process a loan and shorter (50-60) time to approve a loan, respectively, than previously experienced using conventional management processes of assessing applications manually. Additionally, digital banking allows for seamless integration of different banking

services because there will be no data silos and duplicated work that is characteristic of operations of traditional banks.

Digital banking ensures greater efficiency, mainly in back-office processes, services related to customers, and regulations, where there has been significant progress due to digitalization and automation. Processes such as payment processing, account reconciliation, records management, and many other similar stable processes are essential for the back-office, and the error rate can be lowered by 70-80 percent and the process can be accelerated by 60-75 percent because of the possibilities offered by digital banking in this aspect (Furst et al., 2002). With the implementation of chatbots, AI-powered virtual assistants, self-service portals, and able to handle up to 60-80 percent of the routine customer inquiries without requiring any human intervention, efficiency of customer service has improved exponentially, thereby giving the customer service personnel an opportunity to focus on complex cases requiring personal attention (Mbama & Ezepue, 2018). Automation through digitization has made the process of regulatory compliance and reporting easy and efficient, processes that previously required substantial resources, and enabled the automated process of collecting, validating and reporting, which led to savings in compliance costs ranging from 30 to 40 percent (Scott et al., 2017). As highlighted in the findings of a research carried out by Siddik et al. (2016), the digital banking system allows for real-time monitoring and reporting, which help banks to have better control and enhanced risk management efficiency, as well as being proactive in finding solutions to operational constraints. Additionally, digitization ensures faster product development and launch process, and digital banks can launch their products/services within weeks rather than months.

Whereas high operational efficiency is associated with digital banking, its realization could be undermined by factors related to the extent of implementation, preparedness within the organization, and situational opposition such as market characteristics and regulation. Customer-oriented and strategic digital transformation is more efficient than competitive approaches aimed at the implementation of technology without undergoing organizational transformation (Gomber et al., 2018). The efficiency effect is also specific to banks, where large banks benefit from greater efficiency effect compared to smaller banks, considering the economy of scale. However, small banks with a digital focus gain higher relative efficiency effects owing to the removal of legacy system-induced effects (DeYoung et al., 2007). In addition, operational inefficiencies may arise temporarily due to problems related to implementation, including the inability to integrate systems, resistance from employees, and customer resistance due to inefficiency. According to studies by Berger (2003), optimal efficiency benefits from using digital banking services may require at least three to five years of sustained operations. Also, the consistent innovation of digital technologies means there will be constant expenditures for system upgrades, security enhancements, and personnel training to maintain efficiency benefits, creating a dynamic interrelationship between digital

banking advantages and operational efficiency that must be actively managed in line with institutional objectives.

Table 2: Operational Efficiency Improvements Through Digital Banking Implementation

Efficiency Metric	Traditional Banking Performance	Digital Banking Performance	Improvement (%)	Source
Transaction Processing Capacity (transactions/day)	500 - 800	2,000 - 4,000	300% - 500%	Ciciretti et al. (2009)
Transaction Processing Time	2-5 days	2-30 seconds	99.5% - 99.9%	Mbama & Ezepue (2018)
Employee Productivity (revenue per employee)	\$200,000 - \$300,000	\$320,000 - \$480,000	40% - 60%	Hasan et al. (2012)
Transactions per Employee (annual)	8,000 - 12,000	18,000 - 28,000	125% - 233%	DeYoung et al. (2007)
Asset Turnover Ratio	0.04 - 0.06	0.06 - 0.09	50% - 67%	Hernando & Nieto (2007)
Processing Error Rate (%)	2.0% - 3.5%	0.4% - 0.7%	70% - 80% reduction	Furst et al. (2002)
Loan Processing Time	7-14 days	4-6 days	35% - 45%	Kashif et al. (2022)
Loan Approval Cycle Time	10-20 days	4-8 days	50% - 60%	Kashif et al. (2022)
Customer Service Resolution Time	24-48 hours	2-8 hours	67% - 92%	Mbama & Ezepue (2018)
Routine Inquiry Handling (automated %)	10% - 20%	60% - 80%	300% - 400%	Mbama & Ezepue (2018)
Back-office Processing Time	100% (baseline)	25% - 40%	60% - 75%	Furst et al. (2002)

Compliance Reporting Time	5-10 days	1-2 days	70% - 90%	Scott et al. (2017)
Compliance Cost Reduction	100% (baseline)	60% - 70%	30% - 40%	Scott et al. (2017)
Product Launch Time	6-12 months	4-8 weeks	75% - 90%	Siddik et al. (2016)
Overall Operational Efficiency Score (DEA)	0.65 - 0.75	0.82 - 0.92	26% - 42%	Onay & Ozsoz (2013)

Service Quality and Customer Satisfaction

Digital banking has caused a transformation in service quality delivery in the financial service industry as it provides unimaginable convenience, accessibility, and customized customer experience which builds high customer satisfaction levels. Digital banking service quality is generally assessed on modified SERVQUAL dimensions such as reliability, responsiveness, assurance, empathy, and tangibles as well as on digital-specific product features like ease of use, security and availability of the system (Parasuraman et al., 1988; Amin, 2016). Studies show that online banking services can offer high quality service due to 24/7 availability, which removes time and location constraints, which are inherent in traditional banking, provided by branches, thus satisfying the needs and expectations of modern customers in servicing him/hers with instant and one-time banking services (Mbama and Ezepue, 2018). The introduction of easy-to-use interfaces, user-friendly navigation, and customized dashboard has greatly contributed to the user experience, according to studies, 75-85% of digital banking users have reported an increase in the level of satisfaction with their experiences with the new digital banking channel compared to their experiences on the traditional banking channels (Shankar & Jebararakithy, 2019). Moreover, digital banking can support fast-service delivery as most operations were accomplished in minutes, not hours or days, which has an outstanding effect on the aspect of responsiveness of service quality. As Rahi et al. (2019) suggest, the speed and convenience provided by the digital banking platforms has emerged as the primary drivers of customer satisfaction and the time taken to complete the transactions has many strong positive correlations with overall satisfaction indices.

The conceptual links between digital banking and customer satisfaction are mediated by various factors such as system quality, information quality, perceived ease of use, and perceived usefulness as formulated in the Technology Acceptance Model and Information Systems Success Model (DeLone and McLean, 2003). It has been shown in practice that digital banking types that can be rated by a higher customer satisfaction rate by 30-45 per cent than the other digital bank types due to their more reliable and stable

systems with a lower rate of downtime, and also because of their smooth operations (Tam & Oliveira, 2017). The quality of the information, i.e. its accuracy, timeliness, completeness, and relevance that are presented through digital channels affect customer trust and customer satisfaction greatly, and information system quality was rated in the range of 25-35 in terms of customer satisfaction scores improvement (Alalwan et al., 2017). Digital banking technologies, such as personalized product content and customized financial guidance and behavior-based user experiences, have developed into key competitive attributes in customer satisfaction (Kashif et al., 2022). A study conducted by Mbama and Ezepue (2018) has shown that banks that use advanced analytics to provide personalized digital experiences have Net Promoter Scores (NPS) 40-50 points above the standards of banks providing generic digital experiences. Nevertheless, customer satisfaction rates are not affected by digital banking in a homogenous way, and in many cases, there is a substantial difference in this measure across demographic, digital literacy, and the cultural areas, requiring more specific approaches to the delivery of digital service.

Even though, digital banking makes a huge difference in achieving customer satisfaction, in many cases, issues and limitations may have a negative impact on customer satisfaction, which is significantly different across different demographic segments, digital literacy levels, and cultures. Though security and privacy might be the ultimate concerns of the customers 45-60% of the potential digital banking users lack any trust in the quality of such services and are afraid of data breaches, identity theft, and financial fraud, which can greatly undermine the degree of satisfaction despite the high level of the functional service quality (Rahi et al., 2019). The anti-human touch of the digital transactions has left satisfaction gaps with customer segments who appreciate human touch and personalized advisory services, as old customers and high-net-worth customer tend to use the traditional banking channel although there is a digital option (Shankar and Jebarajakirthy, 2019). Customer satisfaction can be greatly diminished as a result of technical issues (such as hard-to-use interfaces, system failures, failed purchases, and poor technical support), and every poor digital experience demands several positive encounters before a customer will be convinced to trust the company again (Alalwan et al., 2017). A study conducted by Tam and Oliveira (2017) shows that service recovery in online banking is especially decisive as dissatisfied consumers can easily switch to the rivals, with a low switching cost, which puts the uniformity of the service quality as a key factor in ensuring that the satisfaction levels remain high. In addition to this, the digital divide still has consequences in the customer perception of the quality of the services with customers who are not digitally literate or do not have access to a decent internet structure being less satisfied and thereby raising equity concerns in the digital banking implementation strategy.

Further to this, the positive effects of digital banking on customer satisfaction have several channels, including and not limited to low costs acquiring customers, higher customer retention rates, greater cross-

selling opportunity, and strengthened brand image, which all contribute to high financial performance. Positive levels of customer satisfaction created by quality digital banking services create a relationship of higher client loyalty and 50-70 retention rates as well as 40-60 purchase intentions than their customers who are not (Mbama & Ezepue, 2018). The cost savings due to higher level of customer satisfaction are high because the cost of acquiring new customers is 5-7 times higher than the cost of maintaining them, and satisfied customers contribute to higher revenues by using products more often and through more word-of-mouth recommendations (Kashif et al., 2022). Banking applications with high-quality services in service quality report 35-50 per cent lower rates of customer complaints, 40-55 per cent fewer queries based on services, and operate operational resources on value-added services and an additional boost in overall bank efficiency (Rahi et al., 2019). Alalwan et al. (2017) state that the banks that are in the top quartile in terms of digital service quality and customer satisfaction have 15-25 percent higher return on equity (ROE) than the industry does, which should reflect direct financial gains of customer-centric digital banking strategies. Nevertheless, optimal customer satisfaction will imply the constant investment in technology enhancements, user experience development, and integration with the omnichannel by which the efficient gains in the sphere of digitalization will not lead to a reduction in the level of service quality.

Table 3: Service Quality and Customer Satisfaction Metrics in Digital Banking

Metric	Traditional Banking	Digital Banking	Improvement (%)	Source
Overall Customer Satisfaction Score (out of 10)	6.2 - 7.0	7.8 - 8.5	21% - 26%	Shankar & Jebarajakirthy (2019)
Service Availability (hours per week)	40 - 50 hours	168 hours (24/7)	236% - 320%	Mbama & Ezepue (2018)
Transaction Completion Time	15-45 minutes	2-5 minutes	78% - 93%	Amin (2016)
Customer Satisfaction with Convenience (%)	55% - 65%	75% - 85%	15% - 31%	Shankar & Jebarajakirthy (2019)
Service Responsiveness Score (1-5 scale)	3.2 - 3.6	4.1 - 4.5	25% - 28%	Rahi et al. (2019)
System Reliability (uptime %)	95% - 97%	99.5% - 99.9%	3% - 5%	Tam & Oliveira (2017)

Net Promoter Score (NPS)	15 - 25	55 - 75	167% - 400%	Mbama & Ezepue (2018)
Customer Retention Rate (%)	65% - 75%	82% - 90%	9% - 38%	Mbama & Ezepue (2018)
Cross-selling Success Rate (%)	12% - 18%	25% - 35%	39% - 192%	Kashif et al. (2022)
Customer Complaint Rate (per 1000 customers)	45 - 65	20 - 30	54% - 69% reduction	Rahi et al. (2019)
Service Quality Score (SERVQUAL, 1-7 scale)	4.8 - 5.4	5.8 - 6.4	18% - 21%	Amin (2016)
Perceived Ease of Use (1-5 scale)	3.0 - 3.5	4.2 - 4.6	31% - 40%	Alalwan et al. (2017)
Customer Trust Level (1-5 scale)	3.5 - 3.9	4.0 - 4.4	13% - 14%	Tam & Oliveira (2017)
Customer Acquisition Cost	\$200 - \$350	\$80 - \$120	57% - 77% reduction	Kashif et al. (2022)
Service-related Query Volume (per month)	100% (baseline)	45% - 60%	40% - 55% reduction	Rahi et al. (2019)
Customer Effort Score (1-7 scale, lower is better)	4.5 - 5.2	2.1 - 2.8	52% - 59%	Shankar & Jebarajakirthy (2019)
Personalization Satisfaction (%)	25% - 35%	65% - 75%	86% - 200%	Mbama & Ezepue (2018)
Security Concern Level (% concerned)	35% - 45%	45% - 60%	-29% to -33% (negative)	Rahi et al. (2019)

Financial Inclusion

Digital banking has become a revolution in promoting financial inclusion through alleviating the obstacles between the formal financial services and extending the banking services to the underserved and unbanked groups, especially the developing economies and the rural communities. Through the digital banking systems, which negate geographical barriers, facilitate the use of lower minimum balance limits, and lower the costs of conducting transactions, financial inclusivity has improved notably since low-income earners can now access formal banking platforms despite lacking formal banking access previously (Demirguc

Kunt et al., 2018). The unbanked have been served especially well by mobile banking and digital wallet to the extent that research has revealed that 1.2 billion adults globally accessed formal financial accounts in 2011 through 2021 due to the digital financial services (Ozili, 2021). The financial inclusion aspects may be digitally-enabled is highly efficient as banks will be able to increase their number of customers and the volume of transactions without corresponding increases in infrastructure expenditures, attaining economies of scale, which enhances the overall institutional efficiency (Siddik et al., 2016). Allen et al. (2016) reported that the digital banking platforms are economically viable to serve customers in remote locations, which is 80-90% lower than the cost of serving customers through the traditional brick-and-mortar concept of business operation, and the economic viability of banks providing services to hitherto uneconomical sections of the market is feasible. Also, electronic banking is used to enable microfinance, low-value transactions that were unprofitable in the models used in traditional banking, increasing the population of addressable markets, and driving upward transaction volume, better data gathering processes, and making sure that risks are better assessed, which, together, lead to better financial performances.

Bank digital workers who are effective at boosting the population formerly avoided by banks see a massive increase in deposits amassed, and previous research indicated that banks that are digitally inclusive grew their deposits by 25-40 percent more than traditional banks that served only the banked population (Gupta & Xia, 2018). Financial inclusion delivers efficiency especially in transaction-related revenue, where the large number of small value-based transactions realized by recently included customer generate significant streams of fee income that can be used to enhance profitability indicators regardless of declining average transaction values (Ozili, 2021). Digital banking platforms permit the banks to capture immense volumes of transaction and behavioural information about the unlike segments of customers, thereby enabling them to formulate alternative credit rating models to enhance the efficiency of lending and minimise the default rates amongst the underserved borrowers (Gabor & Brooks, 2017). A study by Siddik et al. (2016) confirms that Banks that apply financial inclusion programs through the digital channel realize a 15-30% cost-effectiveness ratio improvement and are able to increase the number of customers by 40-80% at the same time. Even so, there is a perceived upside to digital banking because of network effects on promoting financial inclusion and improving bank efficiency, which notwithstanding some serious challenges and obstacles, continues to restrict the realisation of these advantages, especially with digital literacy, cost, and connectivity disparities, regulatory regimes, and lack of trust.

The digital divide is still seen as a decisive limitation, with about 3.7 billion individuals in the global population not having access to the internet or even having limited digital literacy necessary to successfully use digital banking systems to create inclusion gaps that negate the advantages of digital platforms in terms of efficiency (Demirgüç-Kunt et al., 2018). Infrastructure constraints such as poor electricity supply, the

lack of good telecommunications infrastructure, and low smartphone penetration in rural and low-income communities limit the scope of digital banking programs, compelling banks to continue to operate dual channels, making activities inefficient (Ozili, 2021). Distrust and security implications with digitally marginalized groups pose a specific serious problem as such people may not be familiar with formal financial services and digital technology, and surveys show that 40-55% of the lack of use of fintechs is related to a lack of trust in financial institutions or the property right of data privacy (Allen et al., 2016). Operation complexities due to regulatory uncertainty and compliance dictates related to know-your-customer (KYC) practices, anti-money laundering laws, and consumer protection regulation augment efficiency gains of digital inclusion initiatives, especially in regions where identity requirements are high (Gabor and Brooks, 2017). Gupta and Xia (2018) note that the short-term efficiency effects of financial inclusion by banks will be adverse, and meanwhile, the long-term impacts will be positive as it is heavily suggested that banks that effectively incorporate the aim of financial inclusion in their digital banking strategy will attain higher and more sustained efficiency results in the long-term, once the specific institution has reached the critical mass of customer adoption and transaction volume.

More diversified sources of revenue, less risk when it comes to concentration and greater capacity to withstand economic shocks Inclusive models of digital banking balance out the banks in terms of losses of certain demographic or geographic groups of people by serving a larger group of clients (Ozili, 2021). The positive results of efficiency are not restricted to direct financial indicators but also have the social impact dimension such as poverty reduction, economic development, and gender equality which creates positive externalities and increase the social license of banks to operate and may lead to a reduction in regulatory costs and better relationships with stakeholders (Demirgüç-Kunt et al., 2018). Digital financial inclusion banks have remarkably efficient operations, with the most successful digital-inclusive existing in markets such as Kenya, India and Bangladesh having a cost-to-income ratio of 35-45% serving 60-80% previously unbanked customer bases (Siddik et al., 2016). Based on a study conducted by Allen et al. (2016), it is found that the best strategy to differ both inclusion and efficiency is to have platform-based business models which utilize alliances with fintech businesses, technology and telecommunication businesses and retail networks to maximize reach, and reduce operational costs directly. Moreover, the data-driven knowledge created through serving these new, initially invisible consumer groups helps banks to build new products, risk models, and resource allocation in new, more efficient ways that will create sustainable competitive advantages in the market of financial services that is becoming more and more digital.

Table 4: Financial Inclusion Metrics and Efficiency Impacts Through Digital Banking

Metric	Pre-Digital Banking	Post-Digital Banking	Improvement/Change	Source
Access and Reach Metrics				
Account Ownership Rate (% of adults)	51% - 62%	69% - 76%	+18 to +25 percentage points	Demirgüç-Kunt et al. (2018)
Previously Unbanked Population Reached (millions)	Baseline	+1,200 million (2011-2021)	New accounts created	Ozili (2021)
Rural Banking Access (% with access)	25% - 35%	65% - 78%	+40 to +53 percentage points	Siddik et al. (2016)
Cost per Customer in Remote Areas	\$15 - \$20	\$2 - \$4	80% - 90% reduction	Allen et al. (2016)
Minimum Balance Requirement	\$50 - \$100	\$0 - \$5	90% - 100% reduction	Gupta & Xia (2018)
Transaction and Usage Metrics				
Small-value Transaction Viability (minimum)	\$10 - \$25	\$0.10 - \$1.00	96% - 99% reduction	Siddik et al. (2016)
Digital Transaction Volume Growth (%)	5% - 10% annual	35% - 55% annual	+30 to +50 percentage points	Ozili (2021)
Active Account Usage Rate (%)	45% - 55%	72% - 83%	+27 to +38 percentage points	Demirgüç-Kunt et al. (2018)
Mobile Money Account Adoption (% growth)	Baseline	250% - 400%	Substantial expansion	Allen et al. (2016)
Efficiency Impact Metrics				
Customer Base Expansion (%)	Baseline	40% - 80% increase	Through inclusion initiatives	Siddik et al. (2016)
Deposit Growth Rate (annual %)	8% - 12%	20% - 28%	+12 to +20 percentage points	Gupta & Xia (2018)

Cost-to-Income Ratio (inclusive banks)	55% - 65%	35% - 45%	31% - 45% improvement	Siddik et al. (2016)
Cost Efficiency Improvement (%)	Baseline	15% - 30%	Improvement range	Siddik et al. (2016)
Transaction-based Revenue Growth (%)	10% - 15%	35% - 50%	+25 to +40 percentage points	Ozili (2021)
Customer Acquisition Cost (inclusive segment)	\$150 - \$250	\$8 - \$15	88% - 95% reduction	Allen et al. (2016)
Lending and Credit Metrics				
Credit Access for Unbanked (% receiving credit)	9% - 15%	28% - 38%	+19 to +29 percentage points	Gabor & Brooks (2017)
Microfinance Loan Processing Cost	\$25 - \$40	\$3 - \$7	82% - 88% reduction	Gupta & Xia (2018)
Default Rate (alternative credit scoring)	8% - 12%	5% - 8%	33% - 58% reduction	Gabor & Brooks (2017)
Loan Approval Time (small loans)	7 - 14 days	1 - 3 days	71% - 86% reduction	Siddik et al. (2016)
Barriers and Challenges				
Global Internet Access Gap (billions)	N/A	3.7 billion	Remaining excluded	Demirgüç-Kunt et al. (2018)
Digital Literacy Barrier (% citing as concern)	N/A	35% - 48%	Adoption obstacle	Ozili (2021)
Trust/Security Concerns (% of unbanked)	N/A	40% - 55%	Adoption barrier	Allen et al. (2016)
Time to Efficiency Realization (years)	N/A	3 - 5 years	Implementation lag	Gupta & Xia (2018)

Gender Inclusion Impact				
Gender Gap in Account Ownership	9 - 11 percentage points	4 - 6 percentage points	45% - 55% reduction	Demirgüç-Kunt et al. (2018)
Women's Account Ownership (% of adult women)	58% - 63%	72% - 77%	+14 to +19 percentage points	Demirgüç-Kunt et al. (2018)

Discussion

The empirical results of the study show that the effect of digital banking on bank efficiency is significant and diverse with significant improvement realizations in the areas of costs reduction, operations effectiveness, service quality, and financial inclusion. The reduction in the cost outcomes are especially impressive as it was revealed that the cost of transaction processing in digital banking for banks could be reduced to 98.0-98.5, which means that the costs of transferring transactions would fall to \$0.08-0.10 as opposed to 4.00-5.50 (DeYoung et al., 2007). These results can be explained by the theoretical assumption that digital technologies create economies of scale because they allow a bank to work with a greater number of customers without increasing the cost of infrastructures in the same proportion. The cost-to-income improvement of the 15-25 percent gains reported in digitally mature banks (Mbama & Ezepeue, 2018) can be used as a strong indicative of the fact that digital banking has a fundamental change on the banking cost structure that is no longer labor-intensive (not based on fixed costs) but technology-enabled (based on variable costs). Nevertheless, the 3-5 year time between the introduction of digital banking and the achievement of maximum efficiency (Berger, 2003; Gupta and Xia, 2018) clarifies the need to be strategic and persistent and invest money since short-term costs and financial losses may mask long-term gains. The benefits of cost efficiency are not equally shared, and larger institutions are found to reap bigger benefits in terms of absolute savings through economies of scale, whereas smaller digital-native banks realize higher benefits to the costs in terms of relative changes in productivity, capacity of service, and cost efficiency which, on a cumulative basis, improve institutional performance (DeYoung et al., 2007).

The operational efficiency proven in this study does not stop at just instant cash savings, but similar to such savings, shift into significant alterations in productivity, capacity of service and cost efficiency which when asked in total result in changes in performance. The 300-500 percent growth of transaction processing capacity (Ciciretti et al., 2009) as well as 40-60 percent boost in employee productivity in the form of

revenue per employee (Hasan et al., 2012) shows that digital banking produces step-function change and not incremental change in the capacity to process transactions. The argument that these findings have proven is that digital banking is a paradigmatic change to banking business and not just another delivery channel, it is why it has radically changed the way banks make and provide value. The fact that the processing errors were reduced significantly, by 70-80 percent (Furst et al., 2002), and that the processing time of transactions was decreased dramatically, dropping to several seconds (Mbama and Ezepeue, 2018) are objective confirmation that there is an improvement in quality, which is not quantifiable by efficiency indicators, only by the dimensions of reliability and consistency. The extensive character of the gains on the efficiency level, which is determined by the Data Envelopment Analysis (Onay and Ozsoz, 2013), proves the fact that digital banking stimulates the improvement of several input-output relationships at the same time.

Nevertheless, the significant difference in the performance of efficiency among institutions indicates that technology implementation on its own is an insufficient strategy to provide efficiency assistance; instead, organizational aspects such as change management, employee training, and customer adoption strategies play a critical role to moderate the efficiency effects of digital banking investments. This corresponds to the 21-26% of general customer satisfaction scores and a staggering 167-400 percent Net Promoter Scores (Mbama and Ezepeue, 2018; Shankar and Jebarajakirthy, 2019) indicates that digital banking is responding effectively to customer demands to have real-time and in-demand financial services. The 24/7 service and 78-93% drop in time to complete a transaction through its service technologies (Amin, 2016; Mbama and Ezepeue, 2018) cover core customer pain points related to the scope of time and geographical distance in traditional banking. Nevertheless, the security concerns that correspond to 45-60% of customers being concerned with data privacy and fraud (Rahi et al., 2019) can also be seen as a critical vulnerability that can cancel out the efficiency gains in case security breaches destroy customer trust and initiate expensive remediation measures. The directional relationship between efficiency and quality of the service is twofold and intricate: the higher the customer satisfaction, the higher the efficiency in the form of better retention (82-90% vs. 65-75), the success of cross-selling (25-35% vs. 12-18), the lower the costs of handling complaints (Kashif et al., 2022; Mbama and Ezepeue, 2018), which are from the symbiosis and the circles of its improvements caused by quality. However, the reduced levels of satisfaction among older customers along with the desire to engage more in human interaction in some customer groups (Shankar and Jebarajakirthy, 2019) indicates that to maximize efficiency, hybrid models will be more efficient, where digital efficiency is paired with selective human touchpoints to serve high-value or complex interactions.

The financial inclusion results imply that it is not possible to achieve the between-profitability and -inclusion assumptions, given that digital banking opens up new opportunities to pursue social goals in

addition to increasing commercial efficiency. The growth of the accounts owned by 18-25 years old by a percentage of 18.25 and the incorporation of 1.2 billion hitherto unbanked adults across the world (Demirgüç-Kunt et al., 2018; Ozili, 2021) show that digital banking has the potential to transform banking services by democratizing access to these services. Efficiency-wise, the 80-90 percent cost-cutting of the remote customer service (Allen et al., 2016) and the realization of cost-to-income ratios between 35-45 percent serving exclusively unbanked populations (Siddik et al., 2016) are strong signs that operational efficiency and financial inclusion have been a couple of complements and not rivals when digital technologies are being used. The 40-80 percent increase in customer base via inclusion programs, and 25-40 percent growth rates of deposit margins (Gupta and Xia, 2018; Siddik et al., 2016) confirm that the service of previously disqualified groups of the population leads to the emergence of significant commercial prospects that increase the scale economies and lead to more effective efficiency ratios. Nevertheless, it is important to note that the current obstacles such as 3.7 billion individuals without internet connections, the lack of digital literacy, and the lack of trust between the technologically empowered and the under-privileged and under-educated groups (Allen et al., 2016; Demirgüç-Kunt et al., 2018) indicate that the full potential of digitally-enabled inclusion could not be achieved without the secondary measures in terms of infrastructural upgrades, digital literacy, and demonstrations of trustworthiness. The 3-5 years efficiency payback of inclusion-based digital banking initiatives (Gupta and Xia, 2018) suggests that patient capital and long-term strategic commitment is essential, as banks may fail to meet efficiency targets prematurely, misinterpreting these as frightening warning signs to quit their effort to be inclusive and grow to critical mass and sustainability in profitability.

Conclusion

This overall analysis of the effects of digital banking on bank efficiency indicates that digitalization is a paradigm that is revolutionizing the banking business radically and landmark in terms of how the business operates, its cost bases, service delivery and market coverage. The fact is that digital banking has brought significant savings in efficiency levels to many dimensions, such as reduction in transaction processing costs, up to 98% of these costs being dramatic, and operational efficiency, improving by up to 26-42, and customers being more satisfied with scores increasing by 21-26 points, and expanding financial inclusion to 1.2 billion previously unbanked people worldwide. These advantages are very powerful but will be achieved through strategic patience because the time benefit of 3-5 years, large initial investments, and challenges in managing change in an organization can lead to efficiency improvements realized more of a gradual process than an immediate result. The results also display some significant details: the benefits of efficiency increase significantly depending on the size of the bank, the nature of the market, and the quality of implementation; security issues and digital illiteracy diseases remain a limiting factor; and the highest

results are achieved through balanced solutions that combine the digital efficiency of work with the experienced human touchpoints when transactions cannot be automated. However, in the end, digital banking has not only been another delivery channel, but a paradigm shift, and one that would allow banks to reduce costs, improve service quality, and increase market reach among other things, as well as promote the goals of financial inclusion. Nevertheless, long-term success requires ongoing technological advancement, effective cybersecurity practices, design-driven customer focus, and a strategic focus on both business effectiveness and social responsibility agenda. With the banking sector still undergoing digital transformation, the banks that successfully negotiate the implementation issues and remain focused on the needs of the customers and on the operational excellence will gain higher and better competitive advantages in an already digitalised financial service environment.

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